2-3 HISTOGRAMS

- A **histogram** is a compact data summary tool for continuous data, where the data range is divided into specific intervals known as **class intervals**, **cells**, or **bins**. To maximize visual information, **bins** should ideally be of equal width, with a generally satisfactory range of 5 to 20 **bins** based on the square root of the number of observations.

- There is no single rule for determining the number of **bins** in a **histogram**, though various methods and software algorithms exist to guide the process.

  - **No Universal Standard**: There is no officially agreed-upon method for selecting the number of **bins**.

  - **Sturges's Rule**: A common textbook formula used to set the count of **cells**, calculated as $1 + \text{ceiling}(\log_2 n)$.

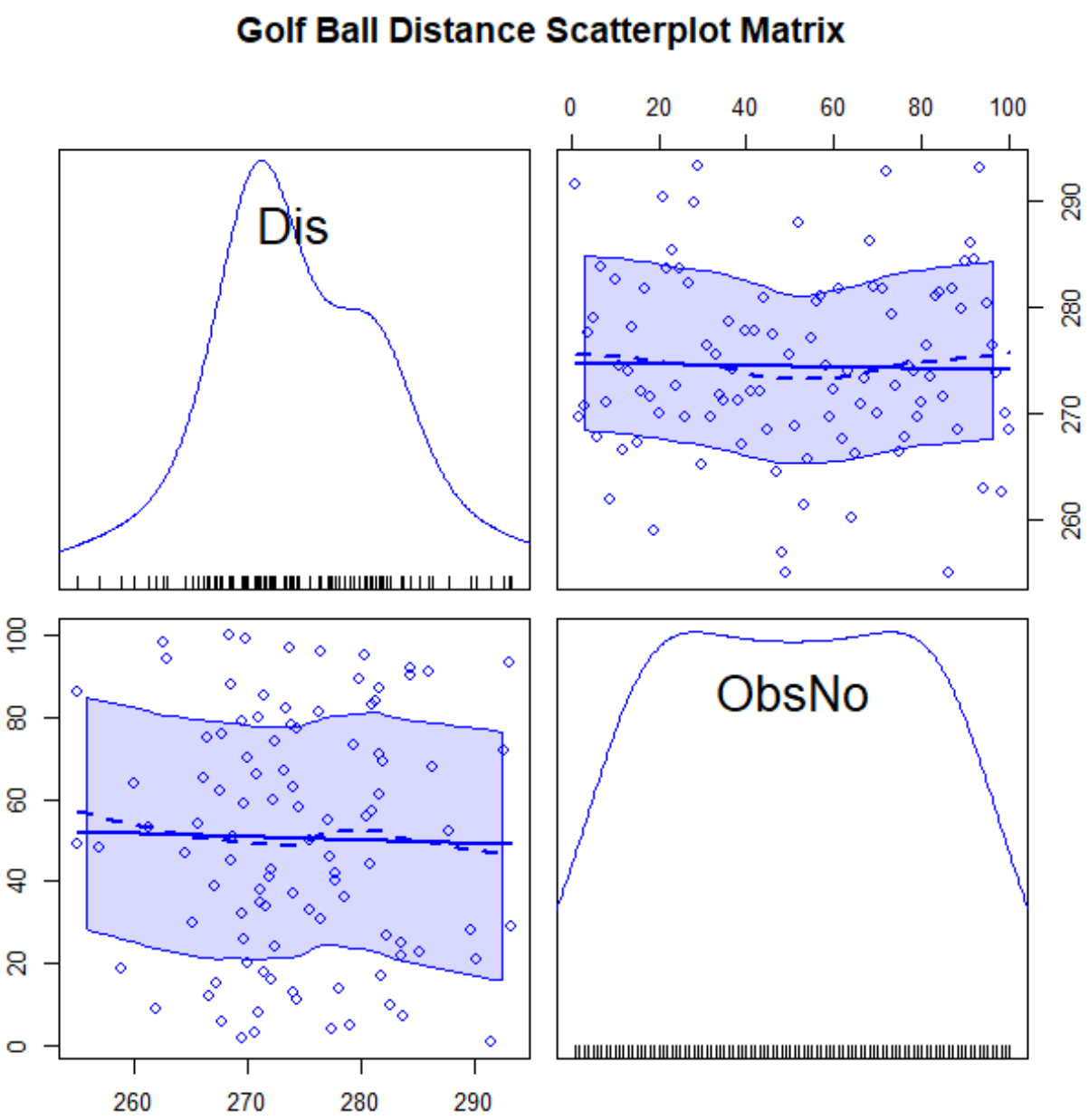
  - **Method Variations**: Numerous versions of Sturges's rule and other mathematical variations are used in practice.

- **Software Diversity**: Programs often use their own unique algorithms to define **class intervals** and widths, which may not align with standard formulas or **quartiles**.

To create a **histogram**, data is sorted into predefined bins, and the count of observations in each bin determines the height of the representing rectangles.

- **Data Sorting**: After defining bin boundaries, observations are counted and assigned to their respective sections.
- **Vertical Scale**: The vertical axis represents the counts, also known as **frequencies**.

- **Proportional Scaling**: If the counts are divided by the total number of observations, the scale represents **relative frequencies**.
- **Visual Representation**: Rectangles are drawn over each bin with heights proportional to either the **frequencies** or **relative frequencies**.
- **Software Use**: Most statistical software packages can automate the construction of these diagrams.


Golf Ball Distance Scatterplot Matrix
Dis
ObsNo
0
20
40
60
80
100
260
270
280
290

- **EXAMPLE 2-6 Golf Ball Distance** involves the USGA using a mechanical device to test the distance performance of 100 golf balls [1]. The data, detailed in **Table 2-4**, is visualized in **Fig. 2-7** via a Minitab histogram featuring 9 active bins and a roughly bell-shaped distribution [1].

  - **Data Source**: **Table 2-4** lists the specific yardage achieved for each of the 100 golf balls tested [1].

  - **Bin Selection**: Based on the square root of the sample size ($\sqrt{100} = 10$), the researchers aimed for 10 bins to ensure a clear visual summary [1].

- **Visual Output**: The **Fig. 2-7** histogram (created via Minitab) displays the data using 9 active bins, starting at a midpoint of 250 yards [1].

- **Distribution Shape**: Much like a stem-and-leaf plot, **Fig. 2-7** reveals that the distances follow a symmetric, bell-shaped distribution [1].

- Histograms are sensitive to bin width, making them best suited for larger data sets containing 75 or more observations. While software often defaults to settings like those in Figure 2-8, these choices are less critical when analyzing larger data sets, such as the 100-observation sample shown in Figure 2-7 and Figure 2-8.

- **Sample Size Recommendation: Histograms are best used with large samples.** They are most effective for **larger data sets** containing at least 75 to 100 observations.

- **Sensitivity**: For small data sets, a histogram's appearance can change drastically if the number or width of bins is adjusted.

- **Default Settings**: **Figure 2-8** shows the result of using a software's default setting, which produced 16 bins for the golf ball data.

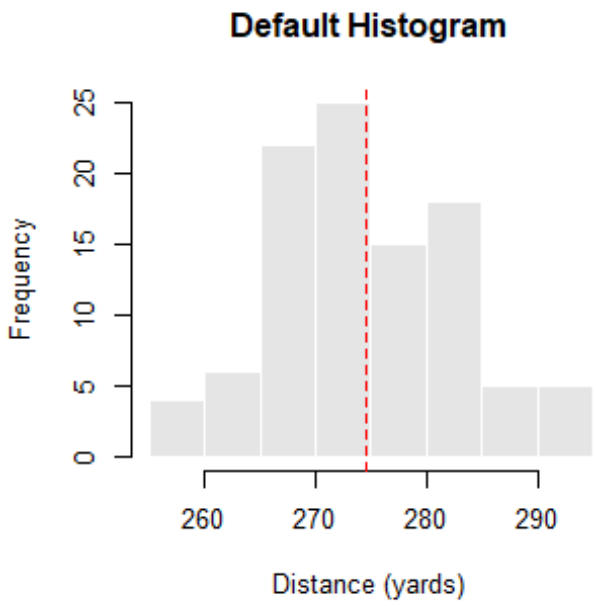
- **Consistency in Large Data**: Because the golf ball sample is large ($n = 100$), the 16-bin version in **Figure 2-8** provides similar insights to the 9-bin version in **Fig. 2-7**.

- **Figure 2-9** displays a **cumulative frequency plot**, a specialized histogram variation where each bar represents the total number of observations falling at or below a specific bin's upper limit.

  - **Cumulative Representation**: The chart shows the running total of data points as they accumulate across bins, rather than individual bin counts.

  - **Data Interpretation**: A **cumulative frequency plot** allows for quick threshold identification, such as determining that roughly 15 out of 100 golf balls exceeded 280 yards.

- **Software Availability**: This visualization is a standard feature in tools like Minitab for analyzing data distributions.


Default Histogram
Frequency
0
5
10
15
20
25
260
270
280
290
Distance (yards)

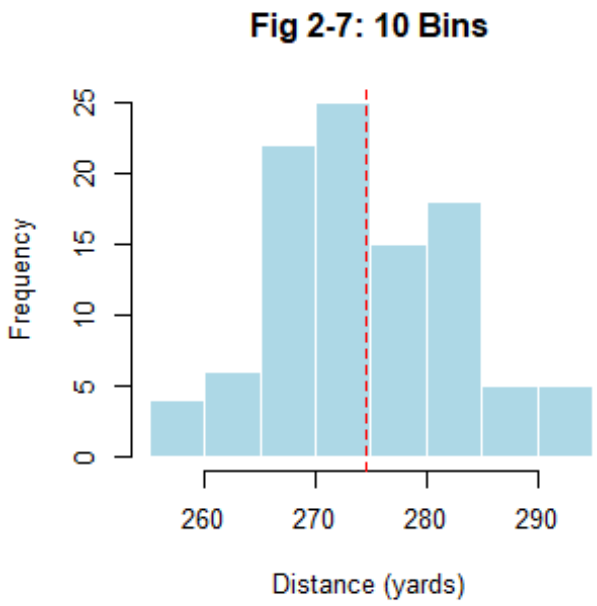


Fig 2-7: 10 Bins
Frequency
0
5
10
15
20
25
260
270
280
290
Distance (yards)

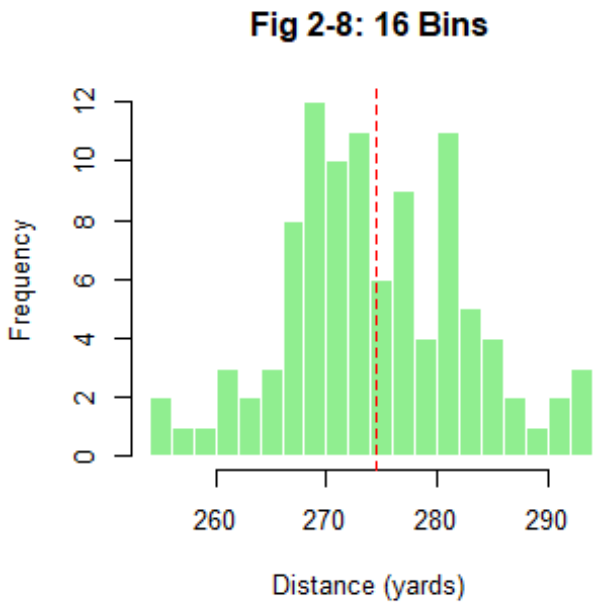


Fig 2-8: 16 Bins
Frequency
0
2
4
6
8
10
12
260
270
280
290
Distance (yards)

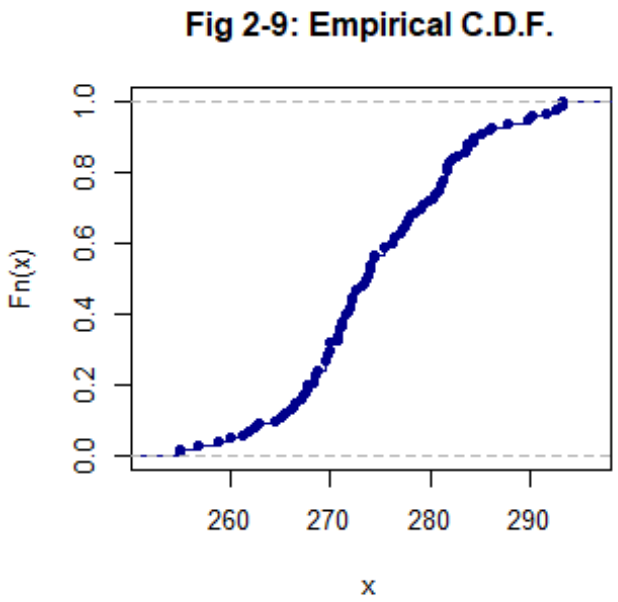
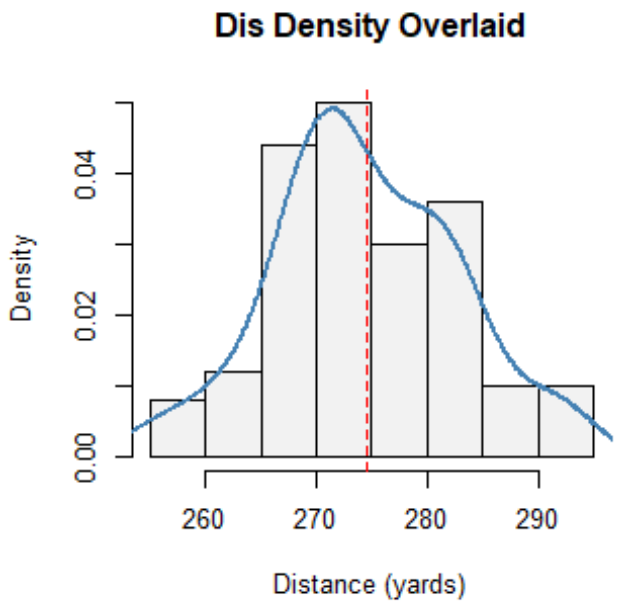
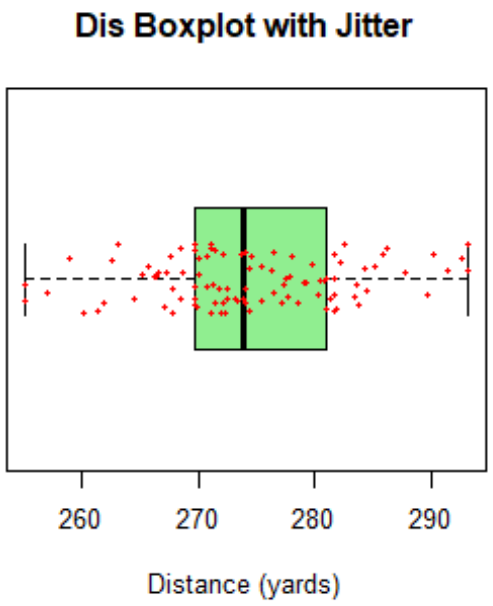
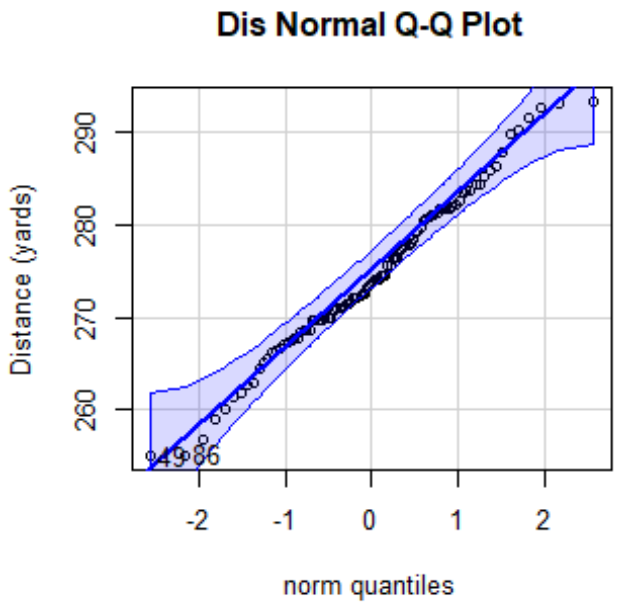


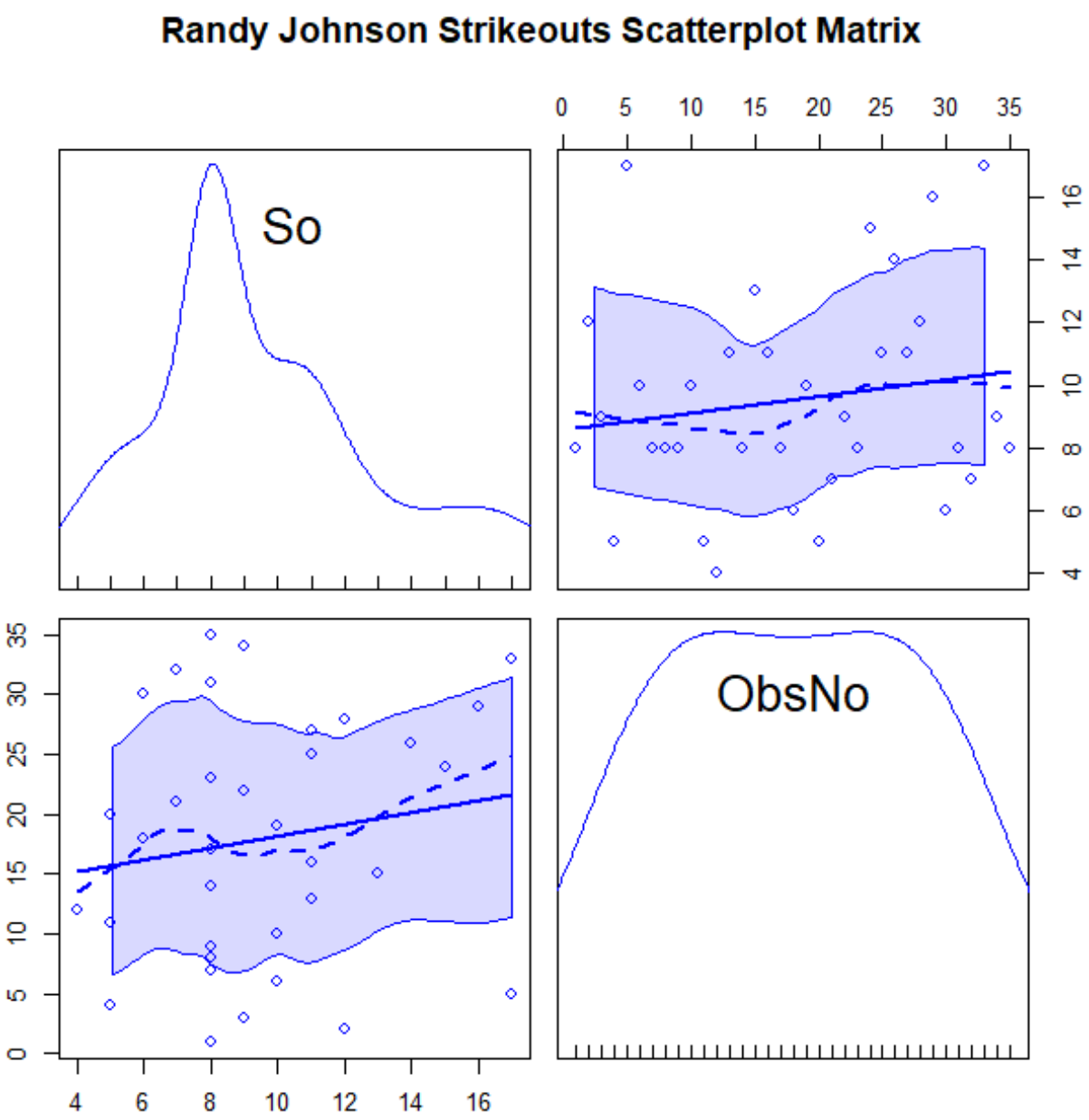
Fig 2-9: Empirical C.D.F.
Fn(x)
0.0
0.2
0.4
0.6
0.8
1.0
260
270
280
290
x


Dis Density Overlaid
Density
0.00
0.02
0.04
260
270
280
290
Distance (yards)


Dis Boxplot with Jitter
260
270
280
290
Distance (yards)


Dis Normal Q-Q Plot
Distance (yards)
260
270
280
290
-2
-1
0
1
2
norm quantiles

- **EXAMPLE 2-7 Randy Johnson** analyzes the 2002 pitching statistics of the Arizona Diamondbacks pitcher to demonstrate the application of **relative frequencies** to discrete data. **Table 2-5** provides a game-by-game performance log for 35 starts, while **Figure 2-10** offers a histogram of strikeout counts, revealing that 42.86% of games had at least 10 strikeouts and 65.71% featured between 8 and 14 strikeouts.


Randy Johnson Strikeouts Scatterplot Matrix
So
ObsNo
0
5
10
15
20
25
30
35
4
6
8
10
12
14
16

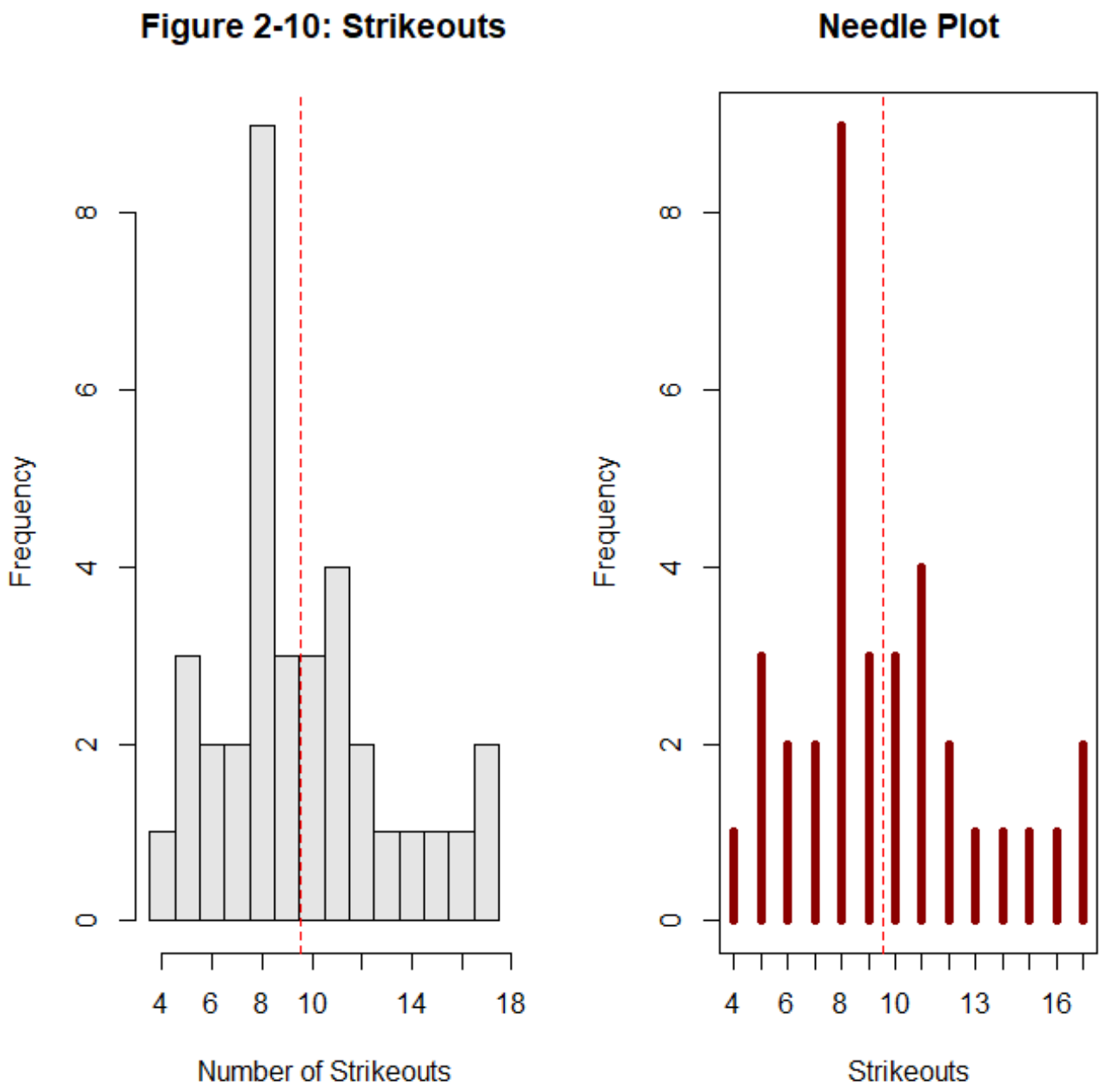


Figure 2-10: Strikeouts
Frequency
0
2
4
6
8
4
6
8
10
14
18
Number of Strikeouts
Needle Plot
Frequency
0
2
4
6
8
4
6
8
10
13
16
Strikeouts

- A **Pareto chart** is a specialized histogram used primarily in quality improvement to identify and prioritize the most significant categories of data, such as defects or failures.

  - **Ordered Categories**: Data is organized by frequency in descending order, with the most common category placed on the far left.

  - **Process Improvement**: Analysts use these charts to focus on "vital" issues by visually distinguishing the most frequent problems from the less significant ones.

- **Pareto's law**: The chart typically demonstrates the principle that a majority of effects (or defects) result from a small number of causes.

- **Origin**: Named after economist Vilfredo Pareto, the chart highlights how a few categories usually account for the bulk of the observations.

- **EXAMPLE 2-8 Aircraft Accidents** uses data from **The Wall Street Journal** (detailed in **Table 2-6**) to analyze hull loss rates across 22 aircraft types between 1959 and 1999. **Figure 2-11** displays a Pareto chart highlighting that three specific aircraft types account for a high percentage of incidents, with an investigation into the **root causes** of the MD-11's high rate [1].

  - **Data Source**: **Table 2-6** lists the total hull losses and the loss rate per million departures, based on an article in **The Wall Street Journal**.

- **Visual Analysis**: **Figure 2-11** shows a Pareto chart, identifying the "vital few" aircraft types with the highest accident rates.
- **Root Causes**: The analysis notes a high accident rate for the MD-11, prompting further investigation into the **root causes**.

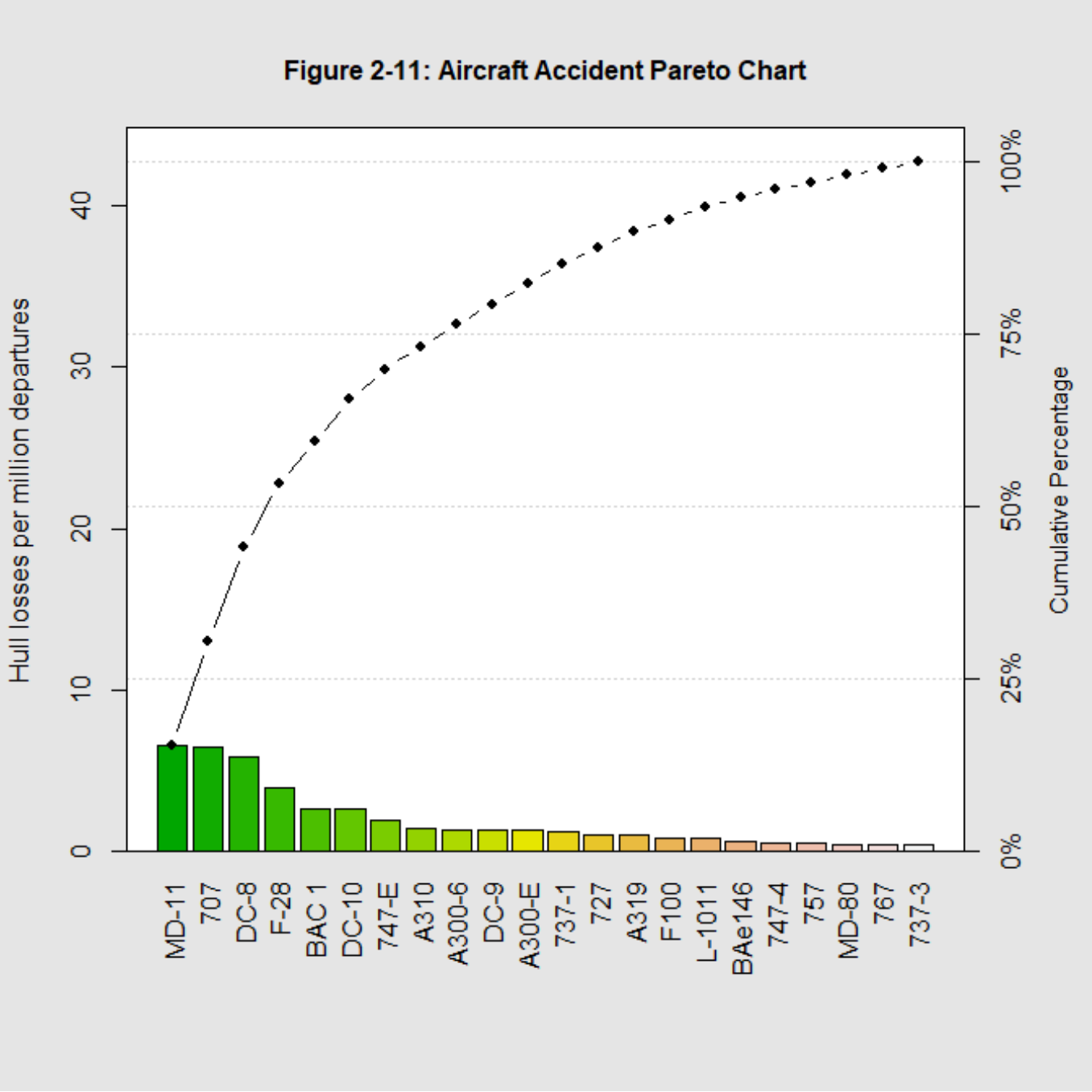


Figure 2-11: Aircraft Accident Pareto Chart
Hull losses per million departures
0
10
20
30
40
Cumulative Percentage
0%
25%
50%
75%
100%
MD-11
707
DC-8
F-28
BAC 1
DC-10
747-E
A310
A300-6
DC-9
A300-E
737-1
727
A319
F100
L-1011
BAe146
747-4
757
MD-80
767
737-3